ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

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Project Name _____
Project Number _____

17.11CP Structural Checklist for Building Type C1: Concrete Moment Frames

Table 17-22. Collapse Prevention Structural Checklist for Building Type C1.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of lines of moment frames in each principal direction is greater than or equal to 2.	5.5.1.1	A.3.1.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COLUMN AXIAL STRESS CHECK: The axial stress caused by unfactored gravity loads in columns subjected to overturning forces because of seismic demands is less than $0.20 f_c^{'}$. Alternatively, the axial stress caused by overturning forces alone, calculated using the Quick Check procedure of Section 4.4.3.6 is less than $0.30 f_c^{'}$.	5.5.2.1.3	A.3.1.4.2	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CONCRETE COLUMNS: All concrete columns are doweled into the foundation with a minimum of four bars.	5.7.3.1	A.5.3.2	

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Moderate Seismicity (Complete the Following Items in Addition to the Items for Low Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of bays of moment frames in each line is greater than or equal to 2.	5.5.1.1	A.3.1.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	INTERFERING WALLS: All concrete and masonry infill walls placed in moment frames are isolated from structural elements.	5.5.2.1.1	A.3.1.2.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COLUMN SHEAR STRESS CHECK: The shear stress in the concrete columns, calculated using the Quick Check procedure of Section 4.4.3.2 is less than the greater of 100 lb/in.^{2} (0.69 MPa) or $2\sqrt{f_c^{'}}$	5.5.2.1.4	A.3.1.4.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	FLAT SLAB FRAMES: The seismic-force-resisting system is not a frame consisting of columns and a flat slab or plate without beams.	5.5.2.3.1	A.3.1.4.3	

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Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
High Seismicity (Complete the Following Items in Addition to the Items for Low and Moderate Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	PRESTRESSED FRAME ELEMENTS: The seismic-force-resisting frames do not include any prestressed or posttensioned elements where the average prestress exceeds the lesser of 700 lb/in.^{2} (4.83 MPa) or $f_c'/6$ at potential hinge locations. The average prestress is calculated in accordance with the Quick Check procedure of Section 4.4.3.8.	5.5.2.3.2	A.3.1.4.4	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CAPTIVE COLUMNS: There are no columns at a level with height-to-depth ratios less than 50% of the nominal height-to-depth ratio of the typical columns at that level.	5.5.2.3.3	A.3.1.4.5	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	NO SHEAR FAILURES: The shear capacity of frame members is able to develop the moment capacity at the ends of the members.	5.5.2.3.4	A.3.1.4.6	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STRONG COLUMN—WEAK BEAM: The sum of the moment capacity of the columns is 20% greater than that of the beams at frame joints.	5.5.2.1.5	A.3.1.4.7	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	BEAM BARS: At least two longitudinal top and two longitudinal bottom bars extend continuously throughout the length of each frame beam. At least 25% of the longitudinal bars provided at the joints for either positive or negative moment are continuous throughout the length of the members.	5.5.2.3.5	A.3.1.4.8	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COLUMN-BAR SPLICES: All column-bar lap splice lengths are greater than $35d_b$ and are enclosed by ties spaced at or less than $8d_b$. Alternatively, column bars are spliced with mechanical couplers with a capacity of at least 1.25 times the nominal yield strength of the spliced bar.	5.5.2.3.6	A.3.1.4.9	

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C	NC	N/A	U			
☐	☐	☐	☐	BEAM-BAR SPLICES: The lap splices or mechanical couplers for longitudinal beam reinforcing are not located within $l_b/4$ of the joints and are not located in the vicinity of potential plastic hinge locations.	5.5.2.3.6	A.3.1.4.10
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	COLUMN-TIE SPACING: Frame columns have ties spaced at or less than $d/4$ throughout their length and at or less than $8d_b$ at all potential plastic hinge locations.	5.5.2.3.7	A.3.1.4.11
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	STIRRUP SPACING: All beams have stirrups spaced at or less than $d/2$ throughout their length. At potential plastic hinge locations, stirrups are spaced at or less than the minimum of $8d_b$ or $d/4$.	5.5.2.3.7	A.3.1.4.12
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	JOINT TRANSVERSE REINFORCING: Beam–column joints have ties spaced at or less than $8d_b$.	5.5.2.3.8	A.3.1.4.13
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	DEFLECTION COMPATIBILITY: Secondary components have the shear capacity to develop the flexural strength of the components.	5.5.2.5.2	A.3.1.6.2
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	FLAT SLABS: Flat slabs or plates not part of the seismic-force-resisting system have continuous bottom steel through the column joints.	5.5.2.5.3	A.3.1.6.3
Diaphragms						
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	DIAPHRAGM CONTINUITY: Floor and roof diaphragms do not have expansion joints or vertical offsets, such as split levels, sawtooth, or clerestory configurations.	5.6.1.1	A.4.1.1
Connections						
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	UPLIFT AT PILE CAPS: Pile caps have top reinforcement, and piles are anchored to the pile caps.	5.7.3.5	A.5.3.8

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